APPENDIX II. ZAMBIA—JOINT IMF/WORLD BANK DEBT SUSTAINABILITY ANALYSIS 2010 UNDER THE DEBT SUSTAINABILITY FRAMEWORK FOR LOW INCOME COUNTRIES

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Based on the Joint Fund-Bank Debt Sustainability Framework for Low-income Countries (LIC DSA), Zambia's risk of debt distress remains low as all debt indicators stay below the indicative policy thresholds throughout the projection period (2010–2030).The public DSA suggests that Zambia's overall public sector debt dynamics are sustainable in light of the current size and the evolution of the domestic debt stock. However, alternative scenarios underscore the need for careful debt management in the face of an expected rise in non-concessional borrowing. Sound macroeconomic policies, including a prudent fiscal position, and strong debt management and project appraisal capacity are needed to maintain debt sustainability, while allowing for non-concessional external borrowing for growth-critical infrastructure investment. The authorities agree with staffs' assessment, highlighting the importance of sound macroeconomic policies and enhanced capacity for debt management and project appraisal.[1]

I. BACKGROUND

1. Zambia is at an important juncture where some of the large-scale projects, including energy and roads, are starting to be financed through non-concessional external borrowing.[2] Zambia's stock of external public debt remains relatively low at 12.4 percent of GPD in 2009, although it has increased slightly since the debt cancellation under the HIPC Initiative and MDRI.[3,4] Most of the public external debt is multilateral (see the table below) and contracted with concessional terms, but the share of non-concessional loans is expected to

[1] The DSA has been produced jointly by Bank and Fund staffs using debt data provided by the authorities.

[2] For example, a non-concessional loan was contracted in 2009 to finance the Kariba North Bank extension project, one of the three major hydropower projects. The development of the Kafue Gorge Lower project is also expected to start in mid-2011 through a public-private partnership.

[3] Zambia's stock of external debt declined substantially from about 86 percent of GDP in 2005 to around 9 percent in 2006. The Government of Zambia continues discussions with the Paris and non-Paris Club members on remaining debt cancellation under the HIPC framework.

[4] The 2009 DSA projected the 2009 debt stock to be US$1,658 million. The actual 2009 debt stock was slightly lower than projected partly owing to shortfalls in the loan disbursement.

increase in 2010/2011 and beyond as the authorities plan to use non-concessional sources to boost the public investment program.

Zambia: External Public and Publicly Guaranteed Debt

	Million USD 2009	% of GDP 1/ 2009
PPG debt	1,587	12.4
Multilateral	1,145	8.9
IMF	348	2.7
Other	797	6.2
Bilateral official	99	0.8
Paris Club	90	0.7
Other	9	0.1
Suppliers and other	133	1.0
Parastatal	210	1.6
Memo item		
GDP USD million	12,805	

Sources: Ministry of Finance and National Planning; and staff estimates
Note: 1/ Figures in percent of GDP may differ in the DSA template due to exchange rates.

2. Zambia's public sector debt also remains low at 24.9 percent of GDP at end 2009.[5] As for the composition, the share of domestic debt is about 57 percent. Treasury bills and bonds comprise 90 percent of domestic debt.

Zambia: Structure of Public Sector Debt

Unit: percent	2009
Public Debt	100.0
External debt	43.2
Domestic debt	56.8

Sources: Ministry of Finance and National Planning and staff estimates

3. This DSA, which is fully consistent with the macroeconomic framework (Box 1), updates the 2009 DSA by incorporating important recent developments in Zambia.[6] Most notably, the current DSA departs from the assumption made in the 2009 DSA that non-concessional external borrowing will moderate in the long run. Rather, this DSA assumes that a sizeable amount of non-concessional borrowing will continue even beyond the medium

[5] With debt relief in 2006 and relatively low budget deficit over the last few years, the domestic financing needs have been lower.

[6] See Appendix II in "Zambia: 2009 Article IV Consultation, Third Review Under the Three-Year Arrangement Under the Poverty Reduction and Growth Facility, and request for Modification of Performance Criteria" IMF Country Report No.10.17, January 2010.

term in light of large infrastructure investment needs.[7] Declining donors' support as a share of GDP also calls for a careful look at the growing needs for alternative sources to finance infrastructure investment and their possible impact on debt sustainability.

II. Underlying DSA Assumptions

4. The Zambia's strong economic performance continues in 2010. GDP is projected to expand by 6.6 percent, supported by vigorous mining exports. Inflation declined to 7.3 percent in October 2010, mainly due to the favorable maize crop. The current account deficit is also expected to narrow, and international reserves have remained solid. The baseline scenario assumes prudent economic policies to foster stable macroeconomic conditions. Inflation is expected to remain in single-digits; real economic growth will continue to be strong supported by commodity exports, infrastructure investment, and expanded electricity generation capacity (Box 1).

5. The baseline scenario also assumes new public external borrowing in the order of US$3.3 billion in 2010-2016 (equivalent to US$ 470 million per year), of which US$1.8 billion is assumed to be on non-concessional terms (54 percent of total loan disbursements).[8] Approximately the same annual amount of disbursements is assumed for the outer years (about US$521 million on average per year) with a share of non-concessional financing remaining at around 54 percent. The average grant element of new borrowing thus declines from 30 percent in 2010 to about 16 percent in 2011–16, remaining at a similar level for the rest of the period.

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[7] A recent World Bank study assesses that Zambia needs to spend an average of $1.6 billion a year over the next decade to develop its infrastructure. See "Zambia's Infrastructure: A Continental Perspective," World Bank, March 2010.

[8] The current Extended Credit Facility program allows for non-zero limits on non-concessional external borrowing: The 2009 program included up to US$400 million of non-concessional loans for electricity projects, and under this limit, US$316 million was contracted in 2009. The 2010 program includes US$600 million of non-concessional debt ceiling for energy and road projects. These limits are on a contracted basis while the DSA accounts for the amount *disbursed* which may not necessarily be the same as the amount *contracted* in a given year.

Box 1: Baseline Macroeconomic Assumptions

Economic growth: Real GDP growth is assumed to pick up to about 7 percent in the medium term supported by mining activity (copper), large infrastructure investment and increased electricity generation capacity. In the long run, the mining sector remains to be important, but economic diversification is also assumed in other areas such as electricity generation, agriculture, and tourism, and the economy is expected to grow at around 6 percent annually.

Inflation: The current objective of monetary policy is to reduce inflation to 8 percent by end-2010 and maintain single digits thereafter. Prudent monetary and fiscal policies are expected to reduce annual inflation to around 5 percent in the medium term.

External sector: Long-term mining output is expected to grow in line with real GDP, and the copper price growth is assumed to stabilize at 3.5 percent in the long run. While copper, whose exports have been particularly strong in 2010, will remain Zambia's most important export, its share in total exports value is expected to decline gradually over time in the medium term and stabilize in the long term (about 65 percent of total exports by 2030 against 76 percent in 2009). Foreign direct investment (FDI) is expected to rise in the medium term, especially in energy and mining sectors, but stabilize in the long run at around 5 percent of GDP. The current account deficit is assumed to worsen to about 6 percent of GDP in the medium term as import-intensive infrastructure investment commences. In the long run, the current account deficit is expected to narrow to around 4 percent of GDP as mining related income outflows stabilize and exports rise as the economy become more diversified.

Government revenue and expenditure: Fiscal policy will be geared at creating space for spending needs to enhance the economic diversification of Zambia. To this end, higher revenue will be mobilized through improvements in tax administration, and new tax policy measures that would broaden tax base. As a result, revenues would rise gradually from 17.8 percent of GDP in 2010 to 23 percent by 2030. Expenditures would stay at an elevated level in the near term mainly due to the growth-critical infrastructure investment, but are expected to remain at about 24 percent of GDP on average over the entire DSA period, with current spending contained.

Government financing: External financing is 0.7 percent of GDP in 2010, expected to increase to about 3 percent of GDP in the near term to finance capital spending with less concessional terms. However, external financing is expected to decline to below 1 percent of GDP over the long term. Domestic financing is 2.6 percent of GDP in 2010 but expected to stabilize at around 0.5 percent of GDP over the rest of the DSA period. A substantial reduction in grant financing is also expected.

III. External Debt Sustainability

Baseline scenario

6. External debt burden indicators rise in the medium term, but they are expected to remain well below the indicative policy thresholds over the projection period (Figure 1 and Table 1).[9] The stock of external public and publicly guaranteed (PPG) debt is expected to rise to around 15 percent of GDP by 2016 but gradually decline thereafter. The present value (PV) of PPG debt to GDP is in the 7 to 14 percent range through 2010-2015, well below the 40 percent threshold, and declines thereafter. The PV of debt to exports ratio increases to 44 percent in 2015, and falls gradually thereafter, remaining well below the threshold of 150 percent. The PV of PPG debt to revenue increases from 41 percent in 2010 to about 70 percent in 2014, falling thereafter, but it is well below the threshold of 250 percent. The debt service to export ratio increases from 1.6 percent in 2010 to 2.8 percent by 2019, but it is also well below the 20 percent threshold.

Stress test

7. The standard sensitivity analysis points to a low risk of debt distress, but the historical scenario highlights potential risks associated with the expected higher level of external borrowing (Table 3). The historical average scenario may not represent a plausible benchmark for Zambia given the significant progress made in the macroeconomic management over the recent years. However, it can also be interpreted as a severe shock scenario where unsound macro policies result in large external imbalances. Under the historical scenario, the PV of debt to GDP ratio approaches the threshold level of 40 percent in 2023, falling back thereafter. This result is mainly driven by the large non-interest current account deficit (10-year historical average of -10.2 percent of GDP), which reflects the poor performance of the Zambian economy in the early 2000s. Similar rising trajectories are observed for other indicators, but they remain below the thresholds. The scenario based on historical averages thus clearly underscores the criticality of strong economic performance supported by sound macroeconomic policies to maintain external debt sustainability in Zambia while allowing for increased non-concessional borrowing.

8. The most extreme stress for the PV of debt-to-GDP ratio and debt-to-revenue ratio is a one-time 30 percent nominal depreciation relative to the baseline in 2011.[10] As a result of this shock, the ratios increase by 33 percent relative to the baseline 2014–15, but remain far below their respective threshold and decline thereafter. The most extreme shock for the PV

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9 The World Bank's Country Policy and Institutions Assessment (CPIA) ranks Zambia as a medium performer (the average 2007–09 CPIA is 3.47). Thus, the external debt burden thresholds for Zambia are (i) a PV of debt-to-GDP at: 40 percent; (ii) a PV of debt-to-exports at: 150 percent; (iii) a PV of debt-to-revenue at: 250 percent; (iv) a debt service-to-exports at: 20 percent: and (v) a debt service-to-revenue at: 30 percent.

10 The most extreme test is defined as the test that yields the highest ratio in 2020.

of debt-to-exports ratio is contracting new public sector loans on less favorable terms in 2010–2030 (i.e. a higher interest rate by 2 percentage points than the baseline). Liquidity indicators remain below their respective thresholds under all tests as well. These results demonstrate that Zambia's risk of debt distress is low even under difficult circumstances with various shocks.

IV. Public Debt Sustainability

Baseline scenario

9. Zambia's public debt rises slightly in the medium term but declines gradually over time. PV of public debt to GDP ratio increases from 24 percent in 2010 to 27 percent by 2014 and comes down thereafter to about 14 percent by 2030. Thus, public debt is rather modest and manageable.

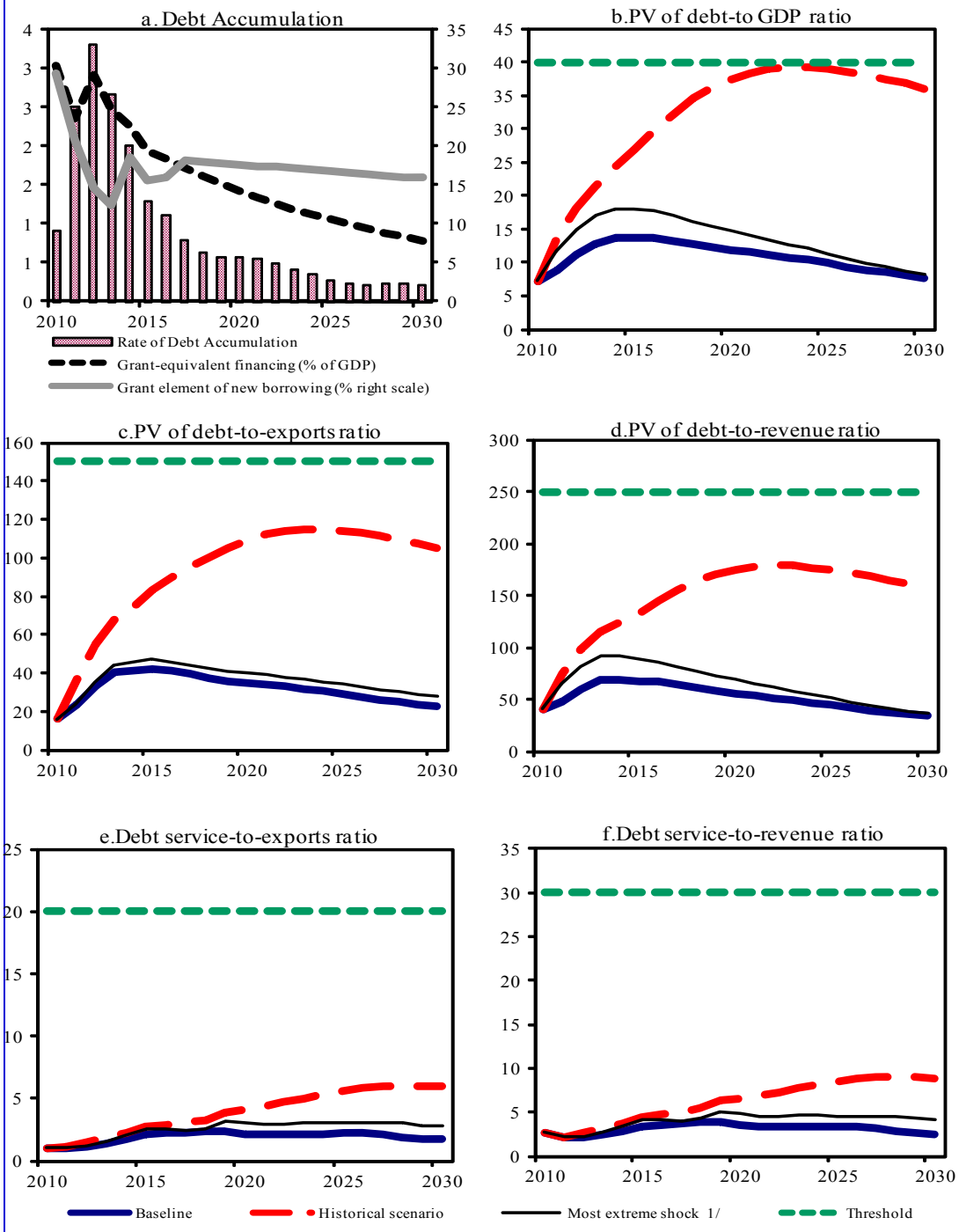
Stress test

10. The standard sensitivity analysis shows that Zambia's public debt is broadly resilient to shocks. Most of the resulting debt indicators under the various standard shocks deviate marginally from those under the baseline and decline over time. However, the results under the fixed primary balance scenario (fixing the primary balance unchanged from 2010) underscore the need for fiscal consolidation in the long run to maintain public debt sustainability (Table 4). Under this scenario, PV of debt-to-GDP ratio and PV of debt-to-revenue ratio keep rising over time, suggesting that the primary deficit in 2010 (1.6 percent of GDP) is not sufficiently small enough to stabilize these indicators. Smaller primary deficit (or even surplus) would be needed to reverse this trend.

V. Debt Distress Classification and Conclusions

11. Staffs recommend that Zambia should be considered at low risk of debt distress based on the results of the external and public DSA. The external debt indicators remain well below their thresholds in the baseline scenario and bound tests. The historical average scenario, however, underlines the importance of sound macroeconomic management to maintain external debt sustainability. The public DSA also suggests that Zambia's overall public sector debt dynamics are sustainable in light of the current size and the evolution of the domestic debt stock in the baseline scenario. On the other hand, the fixed primary balance scenario demonstrates the need for an improved fiscal position in the long run. Finally, in addition to sound macroeconomic policies, strengthening debt management and developing project appraisal capacity are needed in order to maintain debt sustainability in the face of an expected rise in non-concessional borrowing in order to capture the expected growth dividends of such investment. In this regard, the authorities are taking measures to enhance its debt management capacity, including re-organizing the debt management offices in 2011, in line with international best practices.

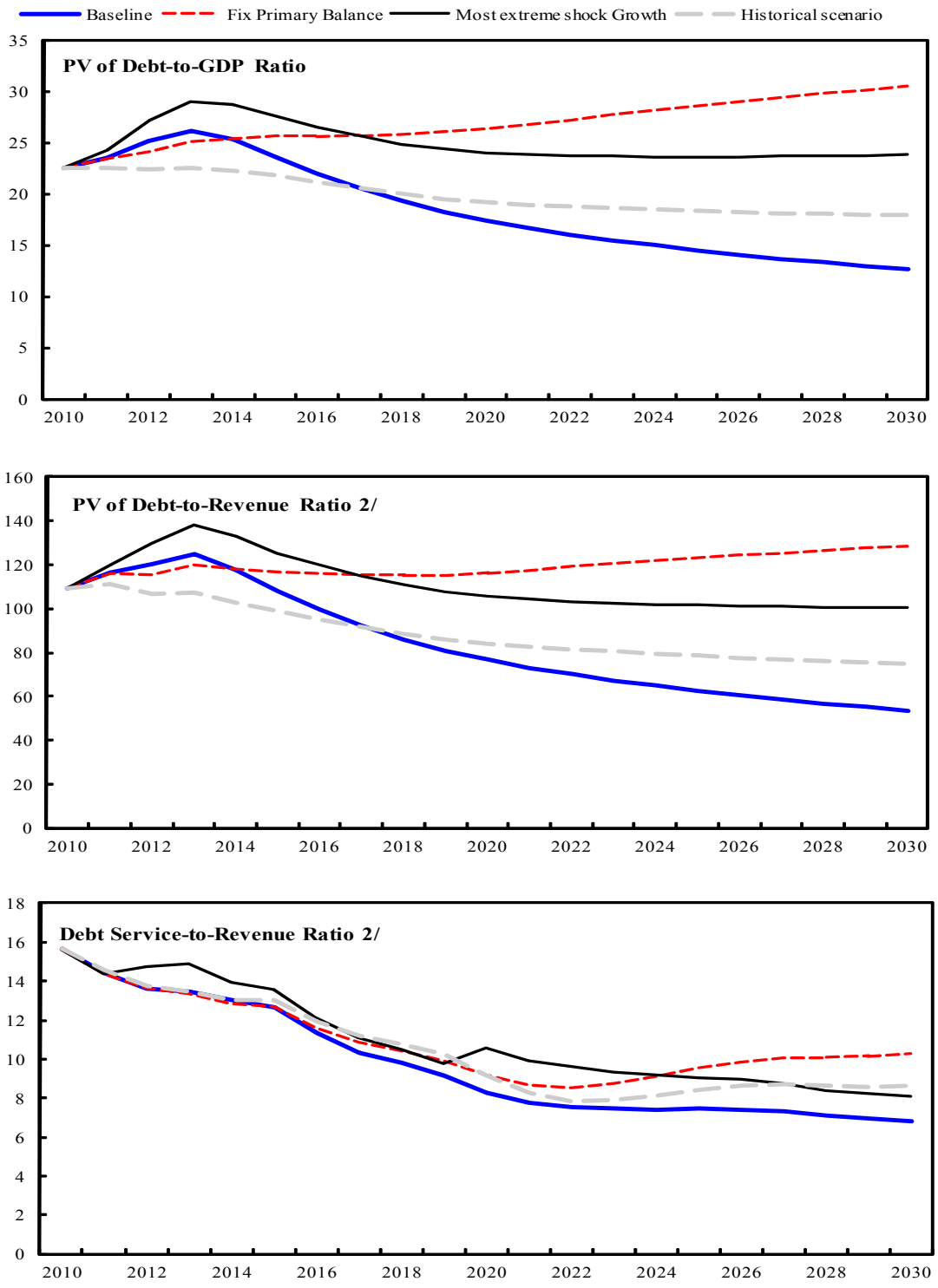
Figure 1. Zambia: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a Terms shock and in figure f. to a Terms shock

Figure 2.Zambia: Indicators of Public Debt Under Alternative Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2020.
2/ Revenues are defined inclusive of grants.

Table 1.: External Debt Sustainability Framework, Baseline Scenario, 2007-2030 1/
(Percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2007	2008	2009			2010	2011	2012	2013	2014	2015	2010-2015 Average	2020	2030	2016-2030 Average
External debt (nominal) 1/	**9.4**	**11.5**	**12.0**			**10.7**	**12.0**	**14.0**	**15.3**	**16.1**	**16.1**		**14.0**	**9.2**	
of which public and publicly guaranteed (PPG)	9.2	11.0	11.4			10.2	11.5	13.6	14.9	15.6	15.6		13.6	8.8	
Change in external debt	0.6	2.1	0.4			-1.3	1.3	2.1	1.3	0.7	0.0		-0.4	-0.4	
Identified net debt-creating flows	-5.4	-1.3	-0.6			-5.3	-2.3	-0.3	-0.4	-1.1	-1.9		-2.5	-1.1	
Non-interest current account deficit	**6.5**	**7.0**	**3.0**	**10.2**	**6.6**	**1.4**	**3.4**	**6.2**	**6.1**	**5.7**	**5.5**		**3.7**	**3.3**	3.6
Deficit in balance of goods and services	-2.2	1.4	-3.4			-9.3	-1.9	2.2	2.9	2.5	2.5		2.1	3.1	
Exports	41.4	35.8	35.6			45.2	37.8	33.1	31.7	32.9	32.3		34.2	34.3	
Imports	39.2	37.2	32.2			35.9	35.9	35.3	34.6	35.4	34.7		36.3	37.4	
Net current transfers (negative = inflow)	-4.6	-3.8	-4.0	-4.4	1.2	-2.8	-2.2	-2.3	-2.2	-2.0	-1.8		-1.3	-0.7	-1.1
of which official	-2.6	-2.2	-2.4			-1.5	-0.9	-1.1	-1.0	-0.9	-0.9		-0.6	-0.3	
Other current account flows (negative = net inflow)	13.3	9.4	10.5			13.5	7.5	6.4	5.4	5.2	4.8		2.8	0.9	
Net FDI (negative = inflow)	**-11.5**	**-6.4**	**-5.5**	**-6.1**	**2.7**	**-6.3**	**-5.2**	**-5.8**	**-5.9**	**-6.1**	**-6.7**		**-5.7**	**-4.0**	-5.3
Endogenous debt dynamics 2/	**-0.4**	**-1.9**	**1.9**			**-0.4**	**-0.5**	**-0.7**	**-0.7**	**-0.7**	**-0.7**		**-0.5**	**-0.3**	
Contribution from nominal interest rate	0.1	0.2	0.2			0.2	0.1	0.1	0.2	0.3	0.4		0.4	0.2	
Contribution from real GDP growth	-0.5	-0.4	-0.8			-0.6	-0.6	-0.7	-0.9	-1.0	-1.1		-0.9	-0.5	
Contribution from price and exchange rate changes	0.0	-1.6	2.5			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**6.0**	**3.4**	**1.0**			**4.0**	**3.5**	**2.4**	**1.7**	**1.8**	**1.9**		**2.1**	**0.6**	
of which exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	8.3			7.8	9.2	11.5	13.2	14.0	14.1		12.4	8.1	
percent of exports	...	...	23.4			17.2	24.3	34.8	41.6	42.6	43.7		36.1	23.6	
PV of PPG external debt	**...**	**...**	**7.8**			**7.3**	**8.8**	**11.1**	**12.7**	**13.6**	**13.7**		**11.9**	**7.8**	
percent of exports	**...**	**...**	**21.9**			**16.1**	**23.2**	**33.5**	**40.2**	**41.3**	**42.4**		**34.9**	**22.6**	
percent of government revenues	**...**	**...**	**48.8**			**40.8**	**47.7**	**60.1**	**68.1**	**69.2**	**67.7**		**55.8**	**33.7**	
Debt service-to-exports ratio (in percent)	**1.4**	**1.9**	**2.0**			**1.6**	**1.5**	**1.7**	**1.9**	**2.2**	**2.6**		**2.6**	**2.1**	
PPG debt service-to-exports ratio (in percent)	**1.2**	**1.2**	**1.2**			**1.1**	**1.0**	**1.2**	**1.4**	**1.7**	**2.1**		**2.2**	**1.7**	
PPG debt service-to-revenue ratio (in percent)	**2.8**	**2.3**	**2.7**			**2.7**	**2.2**	**2.1**	**2.4**	**2.9**	**3.4**		**3.5**	**2.5**	
Total gross financing need (Billions of U.S. dollars)	-0.5	0.2	-0.2			-0.7	-0.2	0.2	0.2	0.1	-0.1		-0.5	0.0	
Non-interest current account deficit that stabilizes debt ratio	5.8	4.9	2.6			2.6	2.1	4.1	4.9	5.0	5.4		4.1	3.7	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.2	5.7	6.3	5.3	1.0	6.6	6.4	6.7	6.8	7.3	7.4	6.9	6.5	6.0	6.3
GDP deflator in US dollar terms (change in percent)	-0.2	20.6	-18.1	10.5	16.8	15.7	10.5	2.1	1.9	2.5	2.6	5.9	3.0	3.0	3.0
Effective interest rate (percent) 5/	0.9	2.1	1.6	1.4	0.6	2.3	1.4	0.6	1.3	2.4	2.6	1.8	2.9	2.5	2.6
Growth of exports of G&S (US dollar terms, in percent)	17.0	9.9	-13.3	20.8	26.4	56.7	-1.7	-4.4	4.2	14.1	8.1	12.8	9.5	9.1	10.0
Growth of imports of G&S (US dollar terms, in percent)	40.4	20.9	-24.7	14.9	17.9	37.8	17.6	7.2	6.8	12.4	8.1	15.0	10.3	9.5	10.1
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	29.5	20.7	14.7	12.5	18.6	15.6	18.6	17.7	15.9	17.0
Government revenues (excluding grants, in percent of GDP)	18.4	18.6	16.0			17.8	18.4	18.5	18.7	19.6	20.2		21.4	23.0	21.9
Aid flows (in Billions of US dollars) 7/	0.7	0.8	0.5			0.5	0.4	0.6	0.5	0.5	0.5		0.6	0.8	
of which Grants	0.5	0.6	0.5			0.4	0.3	0.5	0.5	0.4	0.5		0.5	0.7	
of which Concessional loans	0.2	0.2	0.0			0.0	0.1	0.1	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.0	2.4	2.9	2.5	2.3	1.9		1.4	0.8	1.2
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			82.2	53.7	50.3	52.0	56.3	58.5		62.4	59.4	61.2
Memorandum items:															
Nominal GDP (Billions of US dollars)	11.5	14.7	12.8			15.8	18.6	20.3	22.1	24.3	26.7		43.1	105.1	
Nominal dollar GDP growth	6.0	27.4	-12.9			23.4	17.6	9.0	8.9	10.0	10.1	13.2	9.7	9.3	9.6
PV of PPG external debt (in Billions of US dollars)			1.1			1.2	1.6	2.2	2.8	3.2	3.5		4.7	6.6	
(PVt-PVt-1)/GDPt-1 (in percent)						0.9	2.5	3.3	2.7	2.0	1.3	2.1	0.6	0.2	0.5
Gross workers' remittances (Billions of US dollars)	0.1	0.1	0.0			0.0	0.0	0.0	0.1	0.1	0.1		0.1	0.2	
PV of PPG external debt (in percent of GDP + remittances)	...	...	7.8			7.3	8.7	11.1	12.7	13.5	13.6		11.9	7.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	21.7			16.0	23.0	33.2	39.9	41.0	42.1		34.7	22.5	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	1.2			1.0	1.0	1.1	1.4	1.7	2.1		2.2	1.7	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Zambia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2007-2030
(Percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2007	2008	2009	Average 5/	Standard Deviation 5/	2010	2011	2012	2013	2014	2015	2010-15 Average	2020	2030	2016-30 Average
Public sector debt 1/	25.8	26.5	26.2			25.5	26.3	27.6	28.3	27.3	25.5		19.0	13.7	
of which foreign-currency denominated	9.2	11.0	11.4			10.2	11.5	13.6	14.9	15.6	15.6		13.6	8.8	
Change in public sector debt	-1.6	0.7	-0.3			-0.8	0.8	1.3	0.7	-1.0	-1.8		-0.9	-0.4	
Identified debt-creating flows	-2.4	-0.7	-1.1			-0.7	0.3	1.4	0.7	-0.8	-1.5		-0.9	-0.4	
Primary deficit	-0.4	0.1	1.5	0.2	2.5	1.6	1.8	2.7	1.8	0.3	-0.6	1.3	-0.2	-0.1	-0.2
Revenue and grants	23.0	22.3	20.0			20.6	20.2	20.9	20.9	21.5	21.9		22.6	23.7	
of which: grants	4.6	3.8	4.0			2.8	1.8	2.4	2.2	1.9	1.7		1.2	0.7	
Primary (noninterest) expenditure	22.6	22.4	21.5			22.2	22.0	23.5	22.7	21.7	21.3		22.4	23.6	
Automatic debt dynamics	-1.9	-0.8	-2.6			-2.3	-1.5	-1.3	-1.1	-1.1	-0.9		-0.7	-0.3	
Contribution from interest rate/growth differential	-1.9	-2.0	-1.3			-1.3	-1.3	-1.2	-1.2	-1.1	-1.0		-0.6	-0.3	
of which: contribution from average real interest rate	-0.3	-0.6	0.2			0.3	0.3	0.4	0.6	0.8	0.9		0.6	0.5	
of which: contribution from real GDP growth	-1.6	-1.4	-1.6			-1.6	-1.5	-1.7	-1.8	-1.9	-1.9		-1.2	-0.8	
Contribution from real exchange rate depreciation	-0.1	1.2	-1.2			-1.0	-0.2	-0.1	0.0	0.1	0.1		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.8	1.4	0.8			-0.1	0.5	0.0	-0.1	-0.1	-0.2		0.0	0.0	
Other Sustainability Indicators															
PV of public sector debt	16.6	15.5	22.6			22.5	23.5	25.1	26.1	25.3	23.6		17.3	12.7	
of which foreign-currency denominated	0.0	0.0	7.8			7.3	8.8	11.1	12.7	13.6	13.7		11.9	7.8	
of which external	...	...	7.8			7.3	8.8	11.1	12.7	13.6	13.7		11.9	7.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	6.4	7.2	10.6			10.7	10.8	11.6	10.4	8.6	7.0		4.1	3.6	
PV of public sector debt-to-revenue and grants ratio (percent)	72.2	69.3	113.1			109.5	116.5	120.2	125.1	117.6	107.7		76.6	53.4	
PV of public sector debt-to-revenue ratio (percent)	90.0	83.3	141.8			126.4	128.0	136.0	139.5	128.7	116.6		81.0	55.0	
of which external 3/	…	…	48.8			40.8	47.7	60.1	68.1	69.2	67.7		55.8	33.7	
Debt service-to-revenue and grants ratio (percent) 4/	9.3	13.3	16.3			15.7	14.4	13.6	13.5	13.0	12.7		8.3	6.8	
Debt service-to-revenue ratio (percent) 4/	11.6	16.0	20.4			18.1	15.8	15.4	15.0	14.2	13.7		8.8	7.0	
Primary deficit that stabilizes the debt-to-GDP ratio	1.2	-0.6	1.8			2.4	1.0	1.3	1.2	1.2	1.2		0.7	0.3	
Key macroeconomic and fiscal assumptions															
Real GDP growth (percent)	6.2	5.7	6.3	5.3	1.0	6.6	6.4	6.7	6.8	7.3	7.4	6.9	6.5	6.0	6.3
Average nominal interest rate on forex debt (percent)	0.7	1.8	0.9	1.3	0.6	1.8	0.9	0.1	1.0	2.2	2.3	1.4	2.8	2.5	2.5
Average real interest rate on domestic debt (percent)	-0.5	-3.5	1.9	-3.1	2.7	1.5	2.2	4.4	5.4	5.9	7.4	4.5	7.9	9.2	8.4
Real exchange rate depreciation (percent, + indicates depreciation)	-1.0	13.8	-11.9	-6.5	11.7	-9.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, percent)	10.9	12.8	10.4	18.0	6.1	10.3	7.6	5.5	4.6	4.6	4.5	6.2	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, percent)	0.1	0.0	0.0	0.0	0.2	0.1	0.1	0.1	0.0	0.0	0.1	0.1	0.1	0.1	0.1
Grant element of new external borrowing (percent)	...	...	...	…	…	29.5	20.7	14.7	12.5	18.6	15.6	18.6	17.7	15.9	...

Sources: Country authorities; and staff estimates and projections.
1/ Gross debt of general government and nonfinancial public corporations.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 3. Zambia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030
(Percent)

	Projections							
	2010	2011	2012	2013	2014	2015	**2020**	2030
PV of debt-to GDP ratio								
Baseline	7	9	11	13	14	14	**12**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	7	14	18	21	24	27	**37**	36
A2. New public sector loans on less favorable terms in 2010-2030 2	7	9	12	14	15	15	**14**	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	7	9	11	13	14	14	**11**	6
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	7	10	14	15	16	15	**12**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	7	10	14	16	17	17	**14**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	7	9	13	14	15	15	**12**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	7	6	2	4	5	6	**6**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	7	12	15	17	18.1	18.0	**15**	8
PV of debt-to-exports ratio								
Baseline	16	23	33	40	41	42	**35**	23
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	16	36	55	68	74	84	**109**	105
A2. New public sector loans on less favorable terms in 2010-2030 2	16	24	36	44	46	47	**40**	28
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	16	23	33	39	40	41	**31**	18
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	16	27	43	50	50	50	**38**	19
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	16	23	33	39	40	41	**31**	18
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	16	24	38	45	45	45	**34**	18
B5. Combination of B1-B4 using one-half standard deviation shocks	16	13	4	9	12	13	**13**	13
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	16	23	33	39	40	41	**31**	18
PV of debt-to-revenue ratio								
Baseline	41	48	60	68	69	68	**56**	34
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	41	75	98	114	124	133	**175**	156
A2. New public sector loans on less favorable terms in 2010-2030 2	41	48	64	74	77	75	**64**	42
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	41	48	62	70	70	68	**52**	27
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	41	54	74	80	79	76	**57**	27
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	41	55	76	86	86	83	**65**	34
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	41	50	69	76	75	72	**55**	27
B5. Combination of B1-B4 using one-half standard deviation shocks	41	31	9	21	27	29	**29**	27
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	41	64	81	92	92	89	**69**	36

Table 3. Zambia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030 (continued)
(Percent)

Debt service-to-exports ratio								
Baseline	1	1	1	1	2	2	**2**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	1	1	1	2	2	3	**4**	6
A2. New public sector loans on less favorable terms in 2010-2030 2	1	1	1	2	2	3	**3**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	1	1	1	1	2	2	**2**	2
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	1	1	1	2	2	2	**3**	2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	1	1	1	1	2	2	**2**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	1	1	1	2	2	2	**3**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	1	1	1	0	1	1	**0**	1
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	1	1	1	1	2	2	**2**	2
Debt service-to-revenue ratio								
Baseline	3	2	2	2	3	3	**3**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	3	2	3	3	4	4	**7**	9
A2. New public sector loans on less favorable terms in 2010-2030 2	3	2	2	3	3	4	**5**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	3	2	2	2	3	4	**4**	3
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	3	2	2	3	3	4	**4**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	3	3	3	3	4	4	**4**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	3	2	2	3	3	4	**4**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	3	2	2	1	2	3	**1**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	3	3	3	3	4	5	**5**	3
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	15	15	15	15	15	15	**15**	15

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 4. Zambia: Sensitivity Analysis for Key Indicators of Public Debt 2010-2030

	Projections							
	2010	2011	2012	2013	2014	2015	2020	2030
PV of Debt-to-GDP Ratio								
Baseline	23	24	25	26	25	24	17	13
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	23	22	22	22	22	22	19	18
A2. Primary balance is unchanged from 2010	23	23	24	25	25	26	26	30
A3. Permanently lower GDP growth 1/	23	24	25	26	26	24	19	19
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	23	24	27	29	29	28	24	24
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	23	24	26	27	26	24	18	13
B3. Combination of B1-B2 using one half standard deviation shocks	23	24	25	26	26	25	21	21
B4. One-time 30 percent real depreciation in 2011	23	26	28	29	28	26	20	16
B5. 10 percent of GDP increase in other debt-creating flows in 2011	23	32	33	34	33	31	23	16
PV of Debt-to-Revenue Ratio 2/								
Baseline	109	116	120	125	118	108	77	53
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	109	111	107	107	103	99	84	75
A2. Primary balance is unchanged from 2010	109	116	116	120	118	117	116	129
A3. Permanently lower GDP growth 1/	109	117	121	127	120	111	85	81
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	109	120	130	138	133	125	106	100
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	109	120	124	129	121	111	79	55
B3. Combination of B1-B2 using one half standard deviation shocks	109	117	117	125	121	113	93	87
B4. One-time 30 percent real depreciation in 2011	109	131	133	138	130	120	89	69
B5. 10 percent of GDP increase in other debt-creating flows in 2011	109	158	159	162	152	139	100	66
Debt Service-to-Revenue Ratio 2/								
Baseline	16	14	14	13	13	13	8	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	16	15	14	13	13	13	9	9
A2. Primary balance is unchanged from 2010	16	14	14	13	13	13	9	10
A3. Permanently lower GDP growth 1/	16	14	14	14	13	13	9	8
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	16	15	14	14	14	14	9	9
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	16	14	14	14	13	13	9	7
B3. Combination of B1-B2 using one half standard deviation shocks	16	15	14	14	13	13	9	8
B4. One-time 30 percent real depreciation in 2011	16	15	14	14	14	14	10	10
B5. 10 percent of GDP increase in other debt-creating flows in 2011	16	14	15	15	14	14	11	8

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.